forkcast

Cursor for restaurants

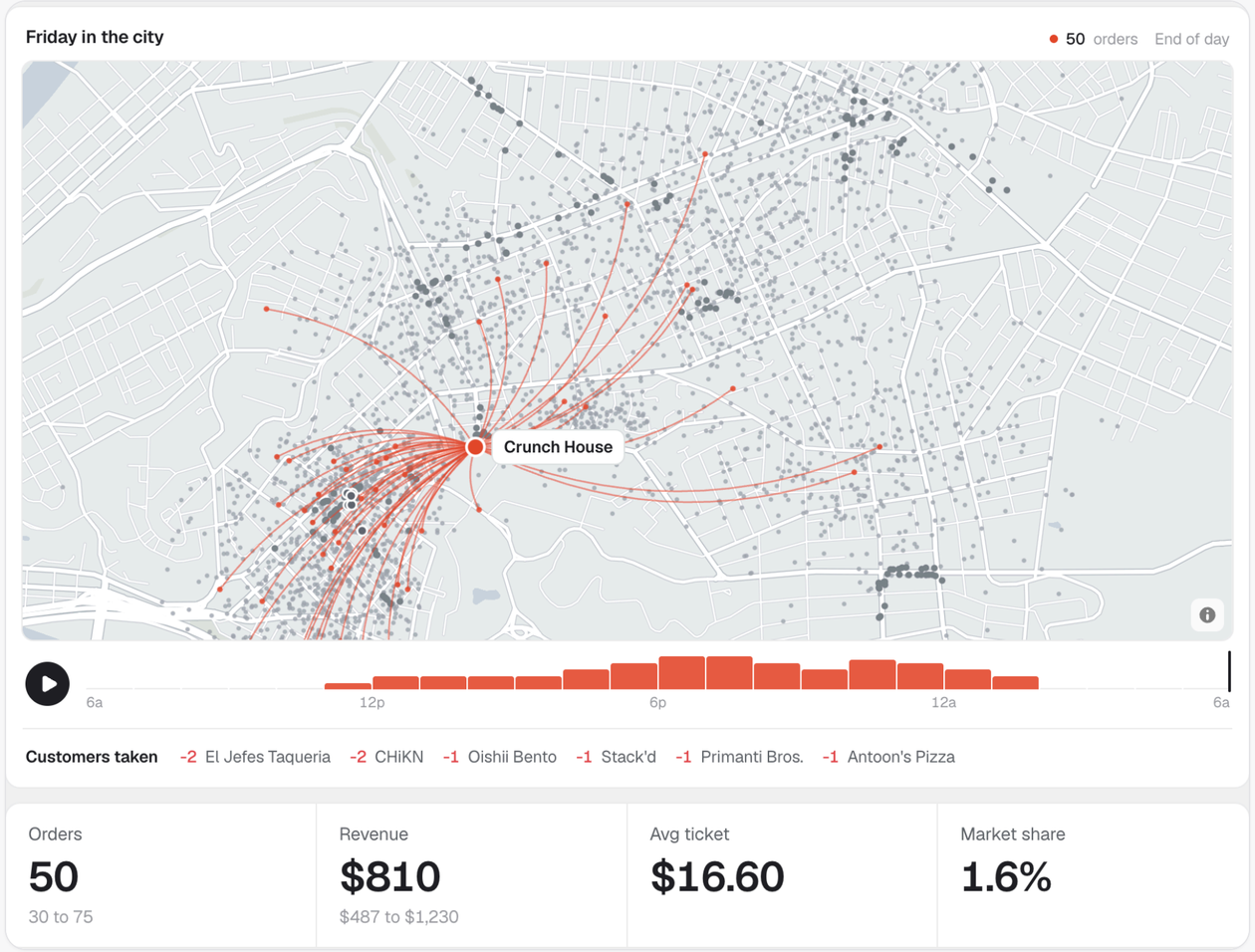
Test a restaurant idea on a simulated city before you open it.

Team Peanuts

Zhiyuan Guo, Evelyn Wang

Track

DoorDash, Back House



A new restaurant's first Friday, before it exists.

The problem

Owners can only test a change by making it

3–5%

typical restaurant margin

Little room for a bad bet.

Source: DoorDash

Decision	How they find out today	If it's wrong
Open a new concept	Sign a lease, open, wait	Savings and a lease
Change prices	Change the menu, watch for weeks	Lost customers or margin
Lower the delivery fee	Run it, check the payouts	Fees given away
Stay open later	Schedule staff, see who comes	Paid hours, no orders

Forkcast lets them try it on a simulated city first.

Who it's for

Two kinds of owners

Opening a new place

Has a concept and a location in mind. Doesn't know who will order, at what hours, or what to charge.

Wants to know

Is there enough demand here, and who would I take customers from?

Already open

Thinking about a lower delivery fee or staying open later. Every real test costs a week of revenue.

Wants to know

Who does the change bring in, and is it worth what it costs?

How it works

01

Describe the idea in chat, or start from a real restaurant

02

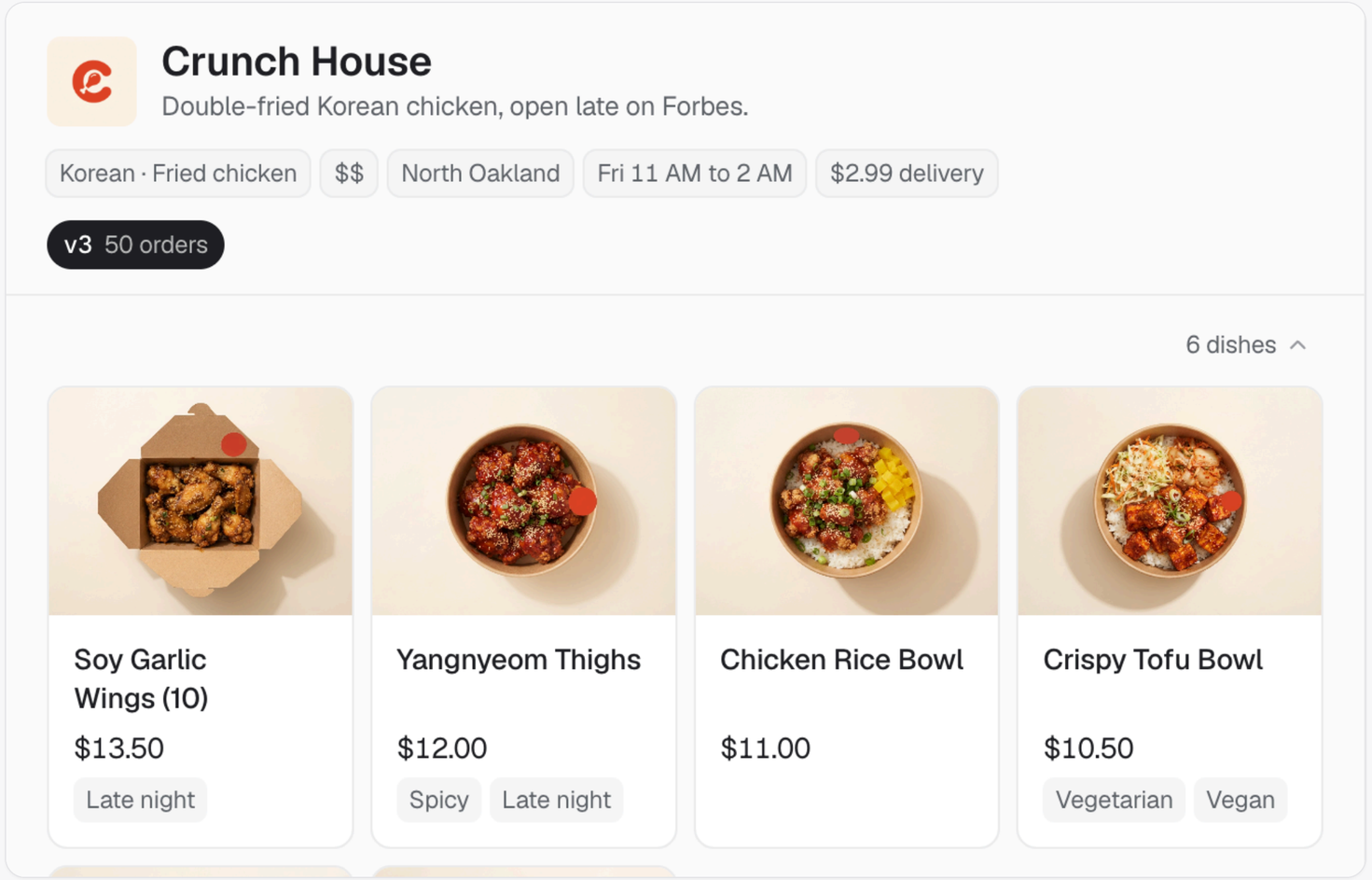
Forkcast drafts it: name, menu, prices, logo, dish photos

03

Simulate a day in the city, hour by hour

04

Set a goal and approve the edits that get there

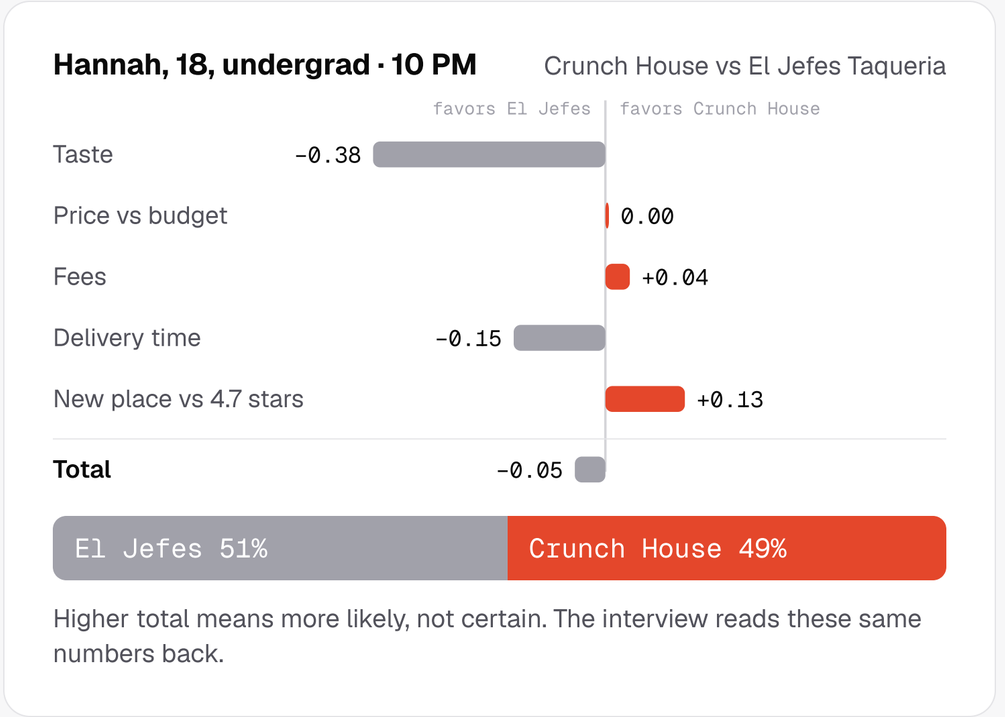


Live demo

What each resident weighs

Each hungry resident scores every open place within 3 km, plus cooking or skipping.

Food	**Taste**	their cuisines, by time of day
	Diet options	needs something they can eat
Money	**Price vs budget**	over budget hurts, under helps a little
	Fees	vs usual $2.99 + 14%, hurt 1.2x more
	Deals	launch discount gets people to try
Speed	**Delivery time**	18 min + 5 min per km
Trust	**Rating**	stars above or below 4.0
	Reviews	more reviews, picked more
	New place	unknown places start behind
	Habit	favorite places get a boost



Higher total means more likely, not certain. The interview reads these same numbers back.

Accuracy

How we know the numbers hold up

Simulator
10,000 residents choose

0.54 vs 0.44
Matches 13 real restaurants it never saw

26/26
Behavior tests pass

Prediction
orders, revenue, why not

100
Simulated Fridays per prediction, so it doesn't wobble

Optimizer
suggests edits

Same days
Before and after see the exact same city

10/10
Edits still win on days it never saw

Owner
decides

Approve
AI suggests, code sets limits, nothing changes without the owner

Accuracy

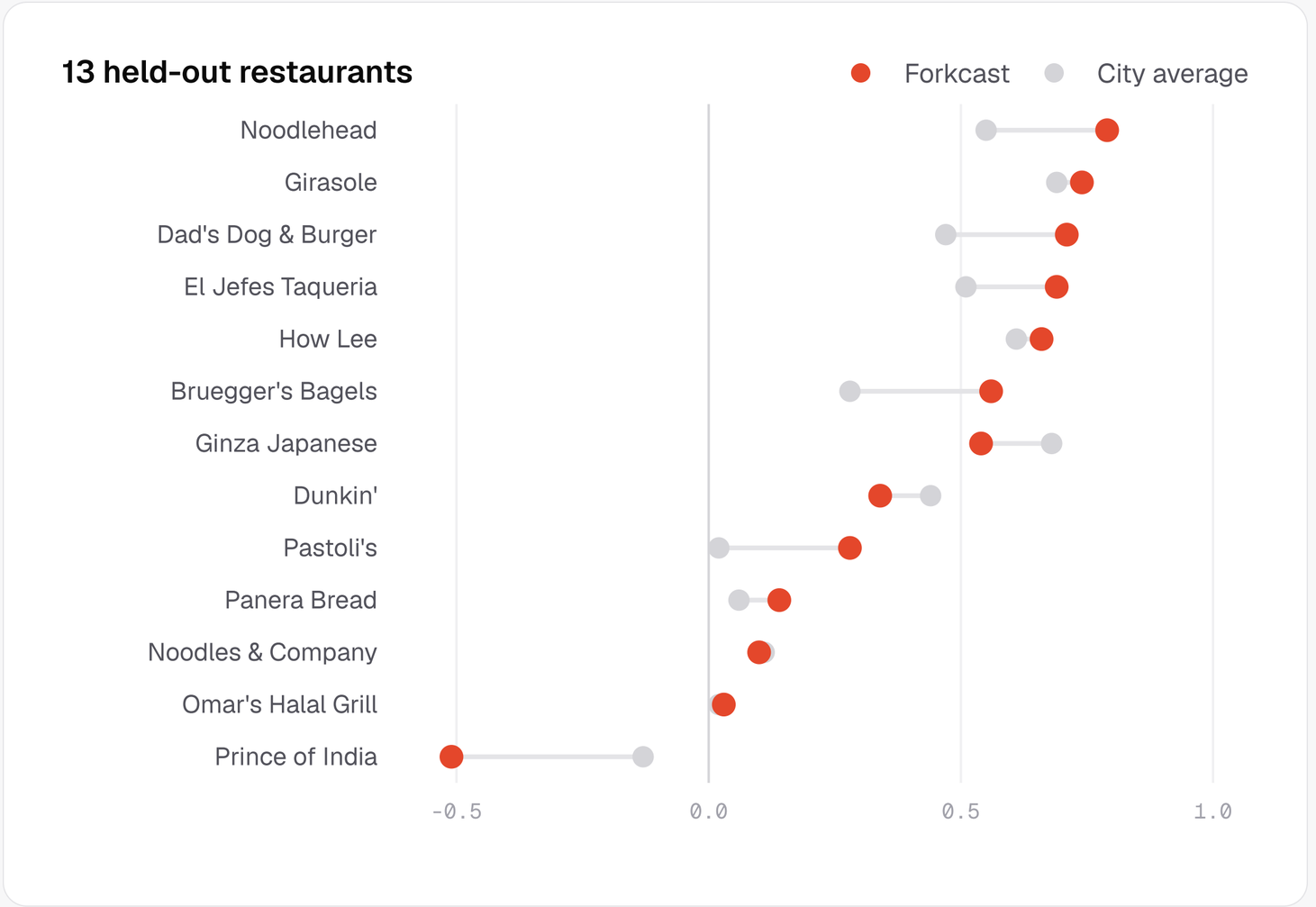
It matches real restaurants it never saw

0.54 Forkcast, calibrated

0.44 City-average curve for every place

0.32 Forkcast before calibration

Median correlation with each place's real hourly curve. We fit on 51 restaurants and tested on 13 the model never saw.



Accuracy

26 checks run on every change

Direction

Cheaper means more orders. Pricier, farther, or closing early means fewer. Free delivery and a launch promo mean more. Monday is below Friday. A vegan main cuts diet objections.

Words can't move numbers

Editing only the tagline gives identical orders.

One story

For sampled residents, the list, the map and the interview agree on what they ordered.

Import a real restaurant unchanged

It should get back its own orders and take nothing from itself.

	Today in the sim	Imported as-is	Taken from itself
Noodlehead	46	45	0
CHiKN	66	69	0
Pizza Palermo	53	55	0
Sushi Fuku	9	10	0

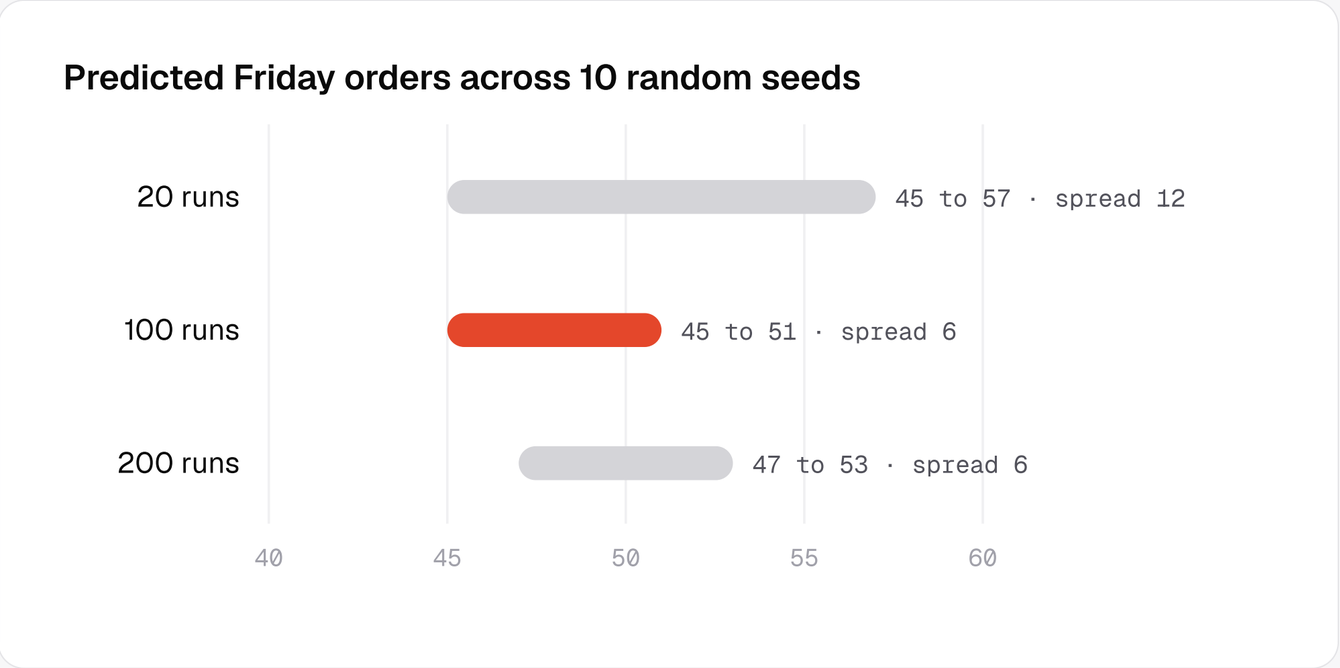
Accuracy

Enough runs that the headline doesn't wobble

Each prediction simulates many Fridays. Every run draws a city-wide demand level, how fast people find a new place, and every resident's orders.

At 20 runs, the headline moved by up to **12 orders** between random seeds. At 100 runs it moves by **6**, and 200 runs doesn't improve on that.

So every prediction uses 100 runs, about 1.4 s.



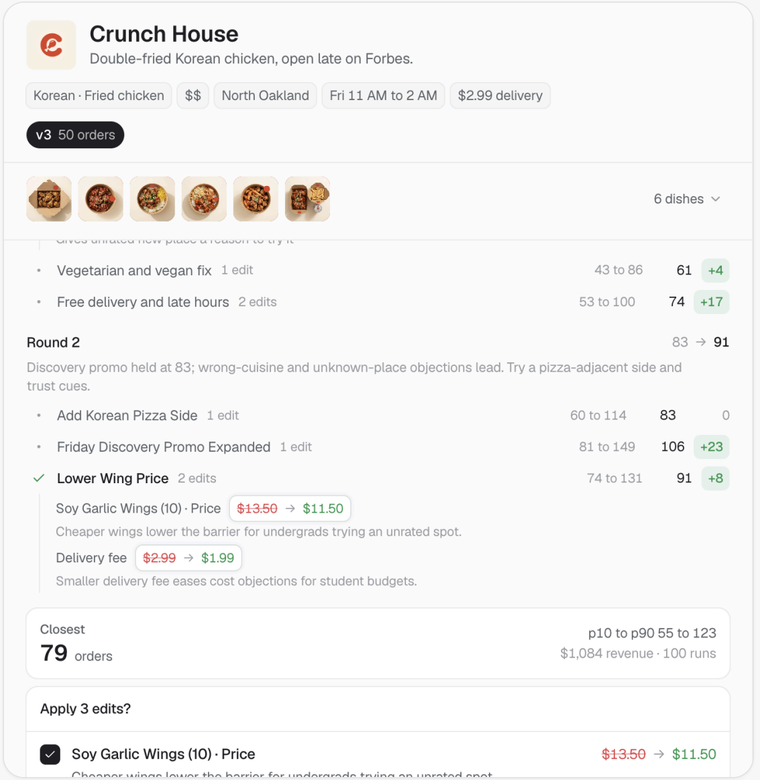
Accuracy

The optimizer is checked on days it never saw

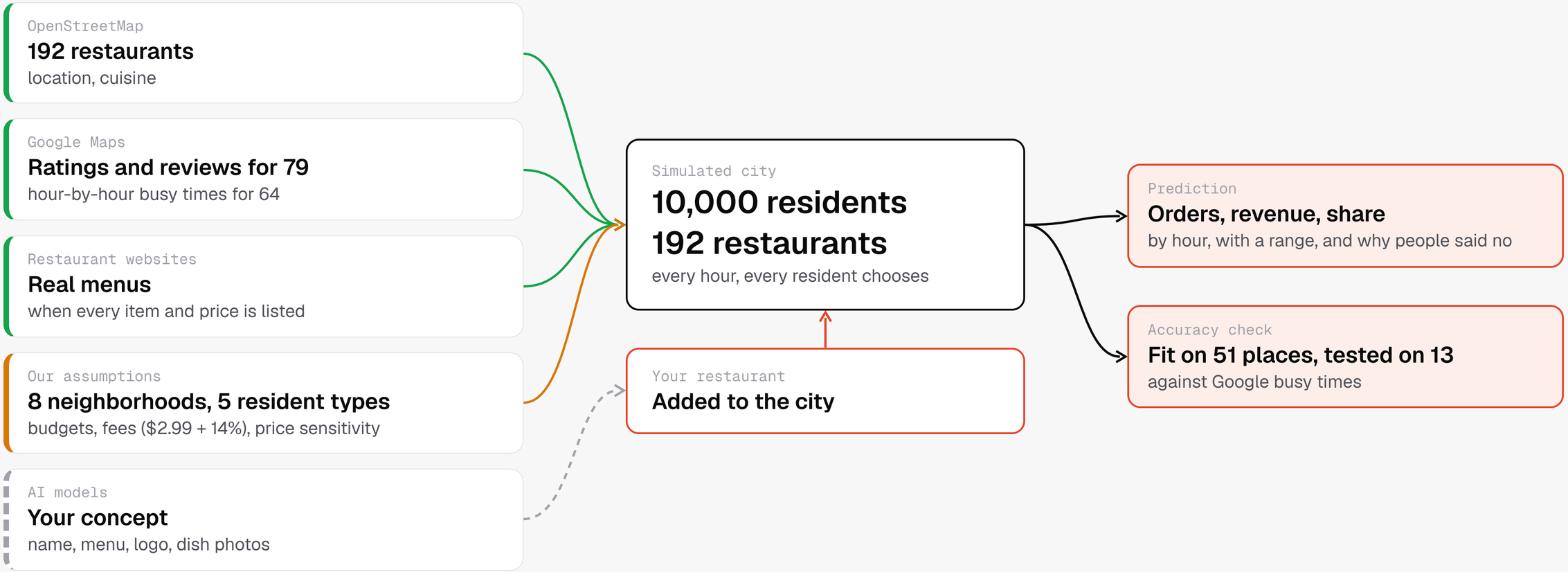
The search tries edit sets and keeps the best score. The best of several noisy scores runs high. So the number the owner sees comes from a fresh check: 100 new Fridays, original and edited concept on the same days.

Search's own score	91 orders
Fresh check, shown to the owner	79 orders
10 more fresh seeds, beat the original	10 of 10
Median across those seeds	49.5 → 83

Goal: 90 orders on Friday. Every edit has limits: 3 per set, no dish under 60% of the owner's price, discounts up to 30%, fee up to $5.99. Nothing applies until the owner approves.



Where the data comes from



Real data · Our assumptions · Generated by AI

Stack TypeScript, Bun, React, MapLibre. Claude for chat, Gemini for images, via OpenRouter.

With DoorDash

What this becomes with real order data

Calibrate on real orders	Swap Google's busy curves for each merchant's order history, so the model learns volume and price sensitivity, not only timing.
Any city	Pittsburgh is built. Another city needs its restaurant list, busy times and resident mix.
Test before you change it	An owner tries a price, hours or fee change in the merchant tools and sees the result before turning it on.

Thanks!

github.com/ZhYGuoL/mise